



Name: _____

Date: _____

SOUND

Solve the following problems. Write your answer on the line.

Score: _____ / 20

Grade 8 | Science | Sheet 3

1) Human ear can hear ____ to 20,000 Hz.	2) Speed of sound in air is about ____ m/s.	3) Musical instruments use ____ to produce sound.	4) Sonar uses ____ to detect underwater objects.
5) Reverberation is prolonged ____.	6) Pitch depends on ____.	7) Loudness depends on ____.	8) Sound is a ____ wave.
9) Compressions and rarefactions are ____.	10) Noise pollution can damage ____.	11) A tuning fork produces a ____ wave.	12) Wavelength is distance between two ____.
13) Speed = frequency x ____.	14) Bats use ____ for navigation.	15) Sonic boom occurs when object moves ____ of sound.	16) Resonance occurs when vibrations ____.
17) Sound energy can be converted to ____ energy.	18) Microphone converts sound to ____ energy.	19) Speaker converts electrical energy to ____ energy.	20) Decibel (dB) scale is ____ (geometric).



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ANSWER KEY

Sound -- Solutions

Grade 8 | Science | Sheet 3

1) Human ear can hear ____ to 20,000 Hz. 20	2) Speed of sound in air is about ____ m/s. 340/343	3) Musical instruments use ____ to produce sound. vibrations	4) Sonar uses ____ to detect underwater objects. ultrasound
5) Reverberation is prolonged ____. sound/echo	6) Pitch depends on ____. frequency	7) Loudness depends on ____. amplitude	8) Sound is a ____ wave. longitudinal
9) Compressions and rarefactions ____. longitudinal	10) Noise pollution can damage ____. hearing/ears	11) A tuning fork produces a ____ tone. pure/single	12) Wavelength is distance between two ____. compressions/rarefactions
13) Speed = frequency x ____. wavelength	14) Bats use ____ for navigation. echolocation	15) Sonic boom occurs when object moves ____ the speed of sound. speed	16) Resonance occurs when vibrations match ____ frequency. natural
17) Sound energy can be converted to ____ energy. electrical	18) Microphone converts sound to ____ signals. electrical signals	19) Speaker converts electrical signals to ____ energy. sound	20) Decibel (dB) scale is ____ (geometric/logarithmic). logarithmic